

Mini-Gemini: Mining the Potential of Multi-Modality Vision Language Models

Siva Prasad M L

Department of Computer Science and Engineering

College of Engineering Munnar

ABSTRACT

The rapid evolution of Large Language Models (LLMs) has spurred significant interest in Vision Language Models (VLMs), which integrate visual and linguistic understanding to handle complex multimodal tasks. Despite notable advances in the field, a pronounced performance gap persists between open academic models and well-established private systems such as GPT-4 and Gemini. Existing VLMs are constrained by insufficient visual resolution, limited data quality, and a lack of unified support for both image comprehension and image generation within a single framework.

This seminar is based on the research paper **Mini-Gemini: Mining the Potential of Multi-Modality Vision Language Models**, which introduces Mini-Gemini, a simple yet highly effective framework designed to systematically mine the latent capabilities of VLMs from three strategic dimensions: efficient high-resolution visual token processing, high-quality multimodal training data, and VLM-guided image generation.

At the architectural core of Mini-Gemini lies a **dual vision encoder** system. A CLIP-pretrained Vision Transformer (ViT) processes low-resolution (LR) inputs to capture global context and long-range patch relationships, while a LAION-pretrained ConvNeXt encoder handles high-resolution (HR) inputs, providing fine-grained spatial detail. These two encoders operate in parallel without increasing the visual token count fed to the LLM.

Bridging the two encoder streams is the proposed **Patch Info Mining** mechanism, which operates through a cross-attention formulation. Low-resolution embeddings serve as query vectors, while high-resolution feature maps provide the keys and values. This design allows the model to selectively extract HR visual detail for each LR patch, enriching the final visual tokens without computational overhead. The framework additionally supports visual token extension to 5N tokens to accommodate higher-resolution inputs when greater detail is required.

To enhance data quality, Mini-Gemini is trained on a curated corpus of approximately 1.2 million image-caption pairs for cross-modality alignment and 1.5 million multimodal conversations for instruction tuning, drawn from publicly available sources including LLaVA, ShareGPT4V, ALLaVA, and OCR-related datasets. A purpose-built 13K instruction-following dataset is further constructed using GPT-4 Turbo to activate the model's **any-to-any generation** capability, enabling it to produce both text and contextually relevant images from a single unified framework via integration with the SDXL diffusion model.

Mini-Gemini supports dense and Mixture-of-Experts (MoE) LLMs ranging from 2B to 34B parameters. Extensive zero-shot evaluations demonstrate that Mini-Gemini achieves leading performance across multiple benchmarks including TextVQA, MMBench, MME, MM-Vet, MMMU, and MathVista. Notably, the 34B configuration with Hermes-2-Yi-34B surpasses private models such as Gemini Pro and

Gwen-VL-Plus on the MMBench and MMMU benchmarks, respectively, demonstrating competitive performance at a fraction of the resource cost.

REFERENCES

- [1] Y. Li, Y. Zhang, C. Wang, Z. Zhong, Y. Chen, R. Chu, S. Liu, and J. Jia, “*Mini-Gemini: Mining the Potential of Multi-modality Vision Language Models*,” arXiv:2403.18814v1 [cs.CV], Mar. 2024. Available: <https://github.com/dvlab-research/MiniGemini>
- [2] H. Liu, C. Li, Y. Li, B. Li, Y. Zhang, S. Shen, and Y. J. Lee, “*LLaVA-NeXT: Improved Reasoning, OCR, and World Knowledge*,” 2024. Available: <https://llava-vl.github.io/blog/2024-01-30-llava-next/>
- [3] Gemini Team, R. Anil, S. Borgeaud et al., “*Gemini: A Family of Highly Capable Multimodal Models*,” arXiv:2312.11805, 2023.